

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f_n(x) \rightarrow f(x)$$

$$e^{ix} = \cos x + i \sin x$$

Chapter 7: Sequences and Series of Functions

Section: Uniform Convergence

Based on Walter Rudin's *Principles of Mathematical Analysis*

$$a_n \rightarrow L$$

$$S_N = \sum_{n=1}^N u_n$$

Outline

Motivation

Uniform Convergence

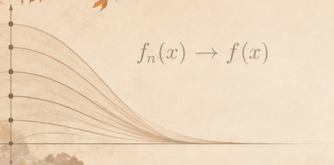
A Test Case: x^n on $[0, 1]$

The Cauchy Criterion

The Supremum Criterion

The Weierstrass M-Test

Summary



A graph showing a family of curves $f_n(x)$ on a coordinate system, all converging towards a limit function $f(x)$. The curves are solid lines, and the limit function is a dashed line. The background is a misty autumn forest with a stream and falling leaves.

$$f_n(x) \rightarrow f(x)$$



A graph showing a sequence of points a_n on a coordinate system, converging towards a limit L . The points are connected by a dashed line, and the limit L is marked on the horizontal axis. The background is a misty autumn forest with a stream and falling leaves.

$$a_n \rightarrow L$$


$$f_n(x) \rightarrow f(x)$$

Motivation

$$\sum_{n=1}^{\infty} u_n(x)$$

$$a_n \rightarrow L$$

$$L$$

Where We Left Off

- Section 7.1: under **pointwise** convergence $f_n \rightarrow f$, continuity, integrals, and derivatives can all be *lost*


$$a_n \rightarrow L$$

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- The culprit: the threshold $N = N(\varepsilon, x)$ may depend on the point x , and be **unboundedly large** near a bad point

Where We Left Off

- Section 7.1: under **pointwise** convergence $f_n \rightarrow f$, continuity, integrals, and derivatives can all be *lost*
- The culprit: the threshold $N = N(\varepsilon, x)$ may depend on the point x , and be **unboundedly large** near a bad point
- The fix: demand **one** N that works for **all** $x \in E$ simultaneously
 $\implies$ *uniform convergence*

Roadmap of This Section

Definition 7.7
uniform convergence

Theorem 7.8
Cauchy criterion

Theorem 7.9
 $M_n = \sup_E |f_n - f| \rightarrow 0$

Theorem 7.10
Weierstrass M-test

Goal: practical tools to **verify** uniform convergence.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f_n(x) \rightarrow f(x)$$

$$a_n \rightarrow L$$

$$S_N = \sum_{n=1}^N u_n$$

Uniform Convergence

Definition 7.7: Uniform Convergence

Definition 7.7

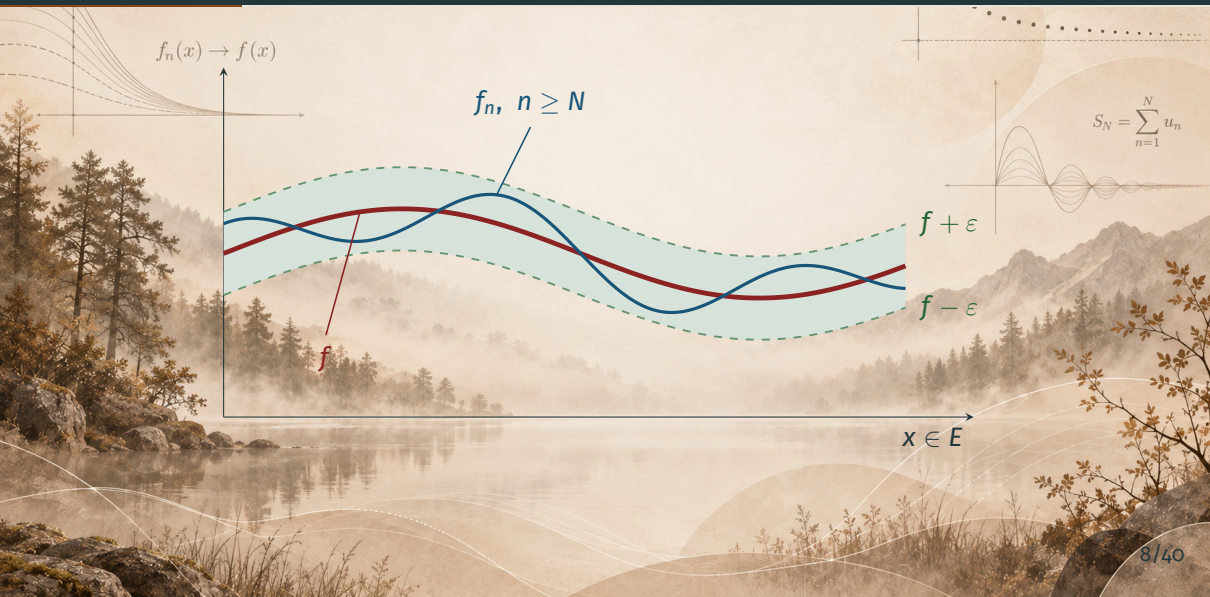
A sequence of functions $\{f_n\}$, $n = 1, 2, 3, \dots$, **converges uniformly** on E to a function f if for every $\varepsilon > 0$ there is an integer N such that $n \geq N$ implies

$$|f_n(x) - f(x)| \leq \varepsilon \quad (12)$$

for **all** $x \in E$.

Key point: N depends on ε *only*, not on x .

Uniform Convergence: The ε -Tube



Pointwise vs. Uniform: Order of Quantifiers

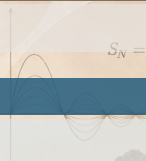
Pointwise on E

$$\forall x \in E \quad \forall \varepsilon > 0 \quad \exists N \quad \forall n \geq N : \quad |f_n(x) - f(x)| \leq \varepsilon$$

Uniform on E

$$\forall \varepsilon > 0 \quad \exists N \quad \forall x \in E \quad \forall n \geq N : \quad |f_n(x) - f(x)| \leq \varepsilon$$

Pointwise: $N = N(\varepsilon, x)$. Uniform: $N = N(\varepsilon)$.


$$S_N = \sum_{n=1}^N u_n$$

Pointwise vs. Uniform: Order of Quantifiers

Pointwise on E

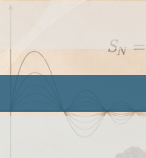
$$\forall x \in E \quad \forall \varepsilon > 0 \quad \exists N \quad \forall n \geq N : \quad |f_n(x) - f(x)| \leq \varepsilon$$

Uniform on E

$$\forall \varepsilon > 0 \quad \exists N \quad \forall x \in E \quad \forall n \geq N : \quad |f_n(x) - f(x)| \leq \varepsilon$$

Pointwise: $N = N(\varepsilon, x)$. Uniform: $N = N(\varepsilon)$.

- Uniform $\implies$ pointwise (same N for every x)
- Pointwise $\nRightarrow$ uniform (example coming up)


$$S_N = \sum_{n=1}^N u_n$$

Uniform Convergence of Series

Definition 7.7 (series)

The series $\sum f_n(x)$ **converges uniformly** on E if the sequence $\{s_n\}$ of partial sums

$$s_n(x) = \sum_{i=1}^n f_i(x)$$

converges uniformly on E .

As always: a series of functions *is* the sequence of its partial sums.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f_n(x) \rightarrow f(x)$$

$$e^{ix} = \cos x + i \sin x$$

$$S_N = \sum_{n=1}^N u_n$$

A Test Case: x^n on $[0, 1]$

Test Case: $f_n(x) = x^n$

Pointwise limit on $[0, 1]$

$$f_n(x) \rightarrow f(x) \quad f_n(x) = x^n \longrightarrow f(x) = \begin{cases} 0, & 0 \leq x < 1, \\ 1, & x = 1. \end{cases}$$

$$S_N = \sum_{n=1}^N u_n$$

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$$S_N = \sum_{n=1}^N u_n$$

- For $0 < x < 1, 0 < \varepsilon < 1$: $x^n \leq \varepsilon \iff n \geq \frac{\log \varepsilon}{\log x} = N(\varepsilon, x) \xrightarrow{x \rightarrow 1^-} \infty$

Test Case: $f_n(x) = x^n$

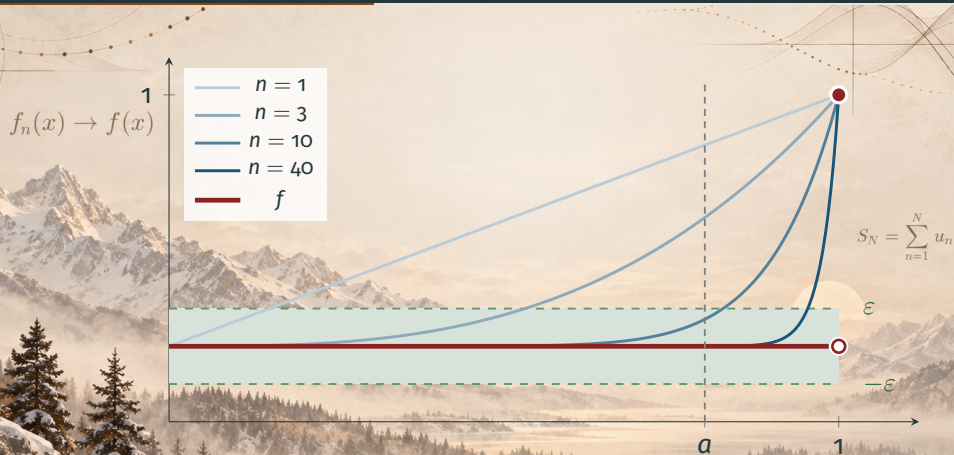
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- For $0 < x < 1, 0 < \varepsilon < 1$: $x^n \leq \varepsilon \iff n \geq \frac{\log \varepsilon}{\log x} = N(\varepsilon, x) \xrightarrow{x \rightarrow 1^-} \infty$
- **Not uniform** on $[0, 1]$: $\sup_{0 \leq x < 1} |x^n - f(x)| = 1$ for every n
- **Uniform** on $[0, a], a < 1$: $x^n \leq a^n \rightarrow 0$ for all $x \in [0, a]$

Test Case: Every x^n Escapes the Tube Near 1



$$S_N = \sum_{n=1}^N u_n$$

$$f_1(x)$$

$$f_2(x)$$

$$f_3(x)$$

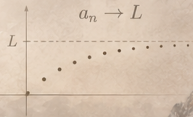
$$\vdots$$

$$f(x)$$

$$f_n(x) \rightarrow f(x)$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

The Cauchy Criterion



Theorem 7.8: Cauchy Criterion for Uniform Convergence

Theorem 7.8 $\rightarrow f(x)$

The sequence of functions $\{f_n\}$, defined on E , converges uniformly on E **if and only if** for every $\varepsilon > 0$ there exists an integer N such that $m \geq N$, $n \geq N$, $x \in E$ implies

$$|f_n(x) - f_m(x)| \leq \varepsilon.$$

$$\sum_{n=0}^{\infty} \frac{1}{n!} = e$$

$$a_n \rightarrow L$$

(13)

Theorem 7.8: Cauchy Criterion for Uniform Convergence

Theorem 7.8

The sequence of functions $\{f_n\}$, defined on E , converges uniformly on E **if and only if** for every $\varepsilon > 0$ there exists an integer N such that $m \geq N$, $n \geq N$, $x \in E$ implies

$$|f_n(x) - f_m(x)| \leq \varepsilon.$$

(13)

Why it is useful: condition (13) mentions only the f_n , **not the limit f** . We can prove uniform convergence *before* we know what f is.

Proof Strategy: Theorem 7.8

Two directions:

$(\Rightarrow)$ Uniform $\Rightarrow$ Cauchy: insert f and use **triangle inequality** with $\varepsilon/2$.

$(\Leftarrow)$ Cauchy $\Rightarrow$ uniform:

(a) each $\{f_n(x)\}$ is a Cauchy sequence of numbers, so it converges.

(Theorem 3.11); call the limit $f(x)$;

(b) let $m \rightarrow \infty$ in (13) to get a bound **uniform in x** .

Proof of Theorem 7.8 — ($\Rightarrow$) Uniform Implies Cauchy

Step 1: Uniform $\Rightarrow$ Cauchy

Let $f_n \rightarrow f$ uniformly on E . Choose N such that $n \geq N, x \in E$ implies

$$|f_n(x) - f(x)| \leq \frac{\varepsilon}{2}.$$

Then for $n \geq N, m \geq N, x \in E$:

$$|f_n(x) - f_m(x)| \leq |f_n(x) - f(x)| + |f(x) - f_m(x)| \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \boxed{\varepsilon}.$$

Proof of Theorem 7.8 — ($\Leftarrow$) Cauchy Implies Uniform

Step 2: The pointwise limit exists

Assume the Cauchy condition (13). For each fixed $x \in E$, $\{f_n(x)\}$ is a **Cauchy sequence** of numbers, so by Theorem 3.11 it converges. Put

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) \quad (x \in E).$$

Thus $f_n \rightarrow f$ pointwise. *Still to show: uniformly.*

Proof of Theorem 7.8 — ($\Leftarrow$) Cauchy Implies Uniform

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Thus $f_n \rightarrow f$ pointwise. *Still to show: uniformly.*

Step 3: Let $m \rightarrow \infty$ in (13)

Given $\varepsilon > 0$, take N from (13). Fix $n \geq N$ and $x \in E$; since $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$,

$$\boxed{|f_n(x) - f(x)| \leq \varepsilon} \quad (n \geq N, x \in E). \quad (14)$$


$$f_n(x) \rightarrow f(x) \\ (n \rightarrow \infty)$$



The Supremum Criterion

$$S_N = \sum_{n=1}^N u_n$$



Theorem 7.9: Uniform Convergence via Suprema

Theorem 7.9

Suppose $\lim_{n \rightarrow \infty} f_n(x) = f(x)$ ($x \in E$). Put

$$M_n = \sup_{x \in E} |f_n(x) - f(x)|.$$

Then $f_n \rightarrow f$ **uniformly** on E if and only if $M_n \rightarrow 0$ as $n \rightarrow \infty$.

$$S_N = \sum_{n=1}^N u_n$$

Theorem 7.9: Uniform Convergence via Suprema

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Proof (immediate from Definition 7.7)

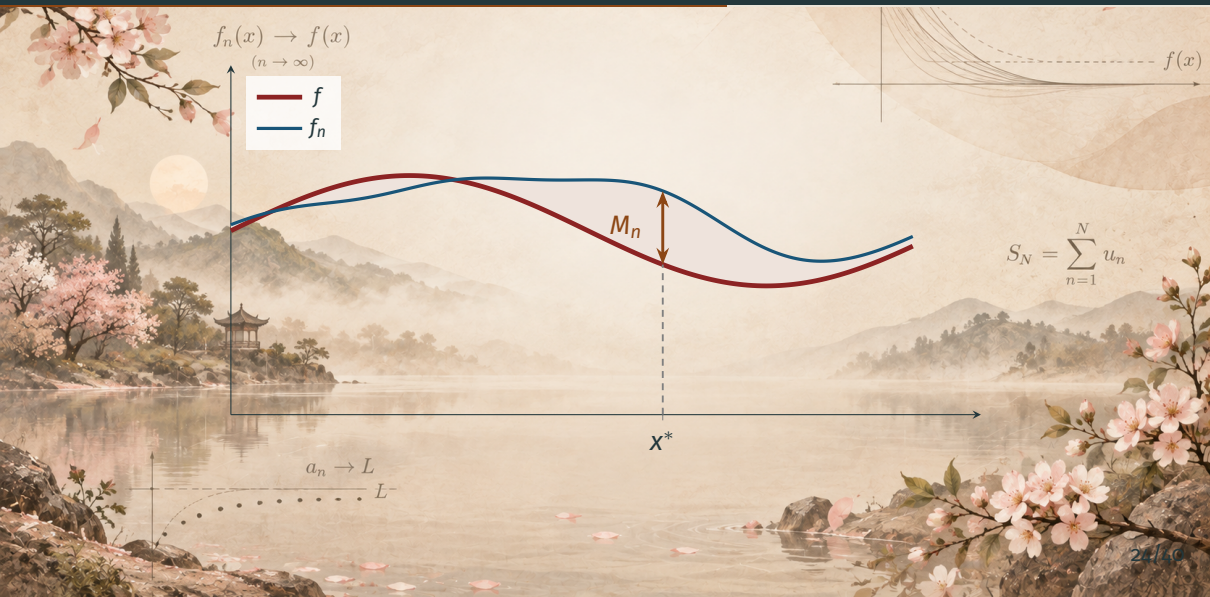
$$|f_n(x) - f(x)| \leq \varepsilon \text{ for all } x \in E \iff M_n \leq \varepsilon$$

(a sup is $\leq \varepsilon$ iff every value is).

$$S_N = \sum_{n=1}^N u_n$$

$$a_n \rightarrow L$$

Theorem 7.9: M_n Is the Largest Vertical Gap



Revisiting the Examples with M_n

$$f_n(x) \xrightarrow{(n \rightarrow \infty)} f(x)$$

Sequence	Limit	M_n	Uniform?
----------	-------	-------	----------

x^n on $[0, 1)$	0	1	no
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$$S_N = \sum_{n=1}^N u_n$$

$$a_n \rightarrow L$$

Revisiting the Examples with M_n

$$\begin{matrix} f_n(x) \rightarrow f(x) \\ (n \rightarrow \infty) \end{matrix}$$

Sequence	Limit	M_n	Uniform?
x^n on $[0, 1)$	0	1	no
Ex. 7.3: $s_n = \sum_{k=0}^n \frac{x^2}{(1+x^2)^k}$ on $\mathbb{R}$	f	$\sup_{x \neq 0} (1+x^2)^{-n} = 1$	no

$$S_N = \sum_{n=1}^N u_n$$

$$\begin{matrix} a_n \rightarrow L \\ L^- \end{matrix}$$

Revisiting the Examples with M_n

$$\begin{matrix} f_n(x) \rightarrow f(x) \\ (n \rightarrow \infty) \end{matrix}$$

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Ex. 7.6: $n^2 x (1-x^2)^n$ on $[0, 1]$	0	$\rightarrow \infty$	no



$$S_N = \sum_{n=1}^N u_n$$

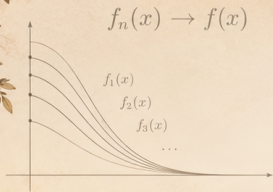


Revisiting the Examples with M_n

$$f_n(x) \xrightarrow{(n \rightarrow \infty)} f(x)$$

Sequence	Limit	M_n	Uniform?
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Ex. 7.3: $s_n = \sum_{k=0}^n \frac{x^2}{(1+x^2)^k}$ on $\mathbb{R}$	f	$\sup_{x \neq 0} (1+x^2)^{-n} = 1$	no
Ex. 7.6: $n^2 x(1-x^2)^n$ on $[0, 1]$	0	$\rightarrow \infty$	no $S_N = \sum_{n=1}^N u_n$
Ex. 7.5: $\frac{\sin nx}{\sqrt{n}}$ on $\mathbb{R}$	0	$\frac{1}{\sqrt{n}} \rightarrow 0$	yes

But: in Ex. 7.5, $f'_n(0) = \sqrt{n} \rightarrow \infty$. Uniform convergence alone does **not** control derivatives.



$$S_N = \sum_{n=1}^N u_n$$



$$\lim_{n \rightarrow \infty} u_n = L$$

The Weierstrass M-Test



Theorem 7.10: The Weierstrass M-Test

Theorem 7.10 (Weierstrass)

Suppose $\{f_n\}$ is a sequence of functions defined on E , and suppose

$$|f_n(x)| \leq M_n \quad (x \in E, n = 1, 2, 3, \dots).$$

Then $\sum f_n$ **converges uniformly** on E if $\sum M_n$ converges.

Here M_n bounds the terms $|f_n|$; it is not the error of Theorem 7.9.

$n=1$

u_n

L

n

$$\lim_{n \rightarrow \infty} u_n = L$$

$$a_n \rightarrow L$$

Theorem 7.10: The Weierstrass M-Test

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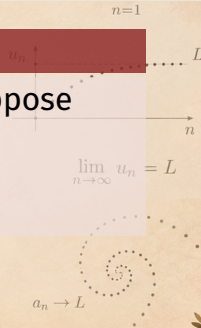
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Then $\sum f_n$ **converges uniformly** on E if $\sum M_n$ converges.

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Remark

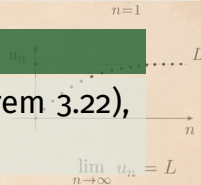
The **converse** is *not* asserted, and in fact it is not true.



Proof of Theorem 7.10

Step 1: Tails of $\sum M_n$ are small

$\sum M_n$ converges, so by the Cauchy criterion for series (Theorem 3.22),
for every $\varepsilon > 0$: $\sum_{i=n}^m M_i \leq \varepsilon$ for all $m \geq n$ large enough.



Proof of Theorem 7.10

Step 1: Tails of $\sum M_n$ are small

$\sum M_n$ converges, so by the Cauchy criterion for series (Theorem 3.22), for every $\varepsilon > 0$: $\sum_{i=n}^m M_i \leq \varepsilon$ for all $m \geq n$ large enough.

Step 2: Transfer to $\sum f_n$ and apply Theorem 7.8

For the same m, n and **every** $x \in E$:

$$|s_m(x) - s_{n-1}(x)| = \left| \sum_{i=n}^m f_i(x) \right| \leq \sum_{i=n}^m |f_i(x)| \leq \sum_{i=n}^m M_i \leq \varepsilon.$$

So $\{s_n\}$ satisfies (13); by Theorem 7.8 it converges uniformly.

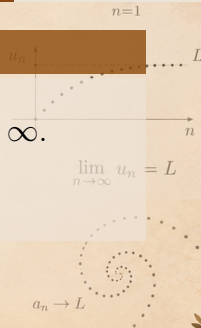
Example: A Trigonometric Series

Applying the M-test

$$\sum_{n=1}^{\infty} \frac{\sin nx}{n^2}, \quad \left| \frac{\sin nx}{n^2} \right| \leq \frac{1}{n^2} = M_n \quad (x \in \mathbb{R}),$$

$$\sum \frac{1}{n^2} < \infty.$$

$\Rightarrow$ the series converges **uniformly on all of $\mathbb{R}$.**



Example: A Trigonometric Series

Applying the M-test

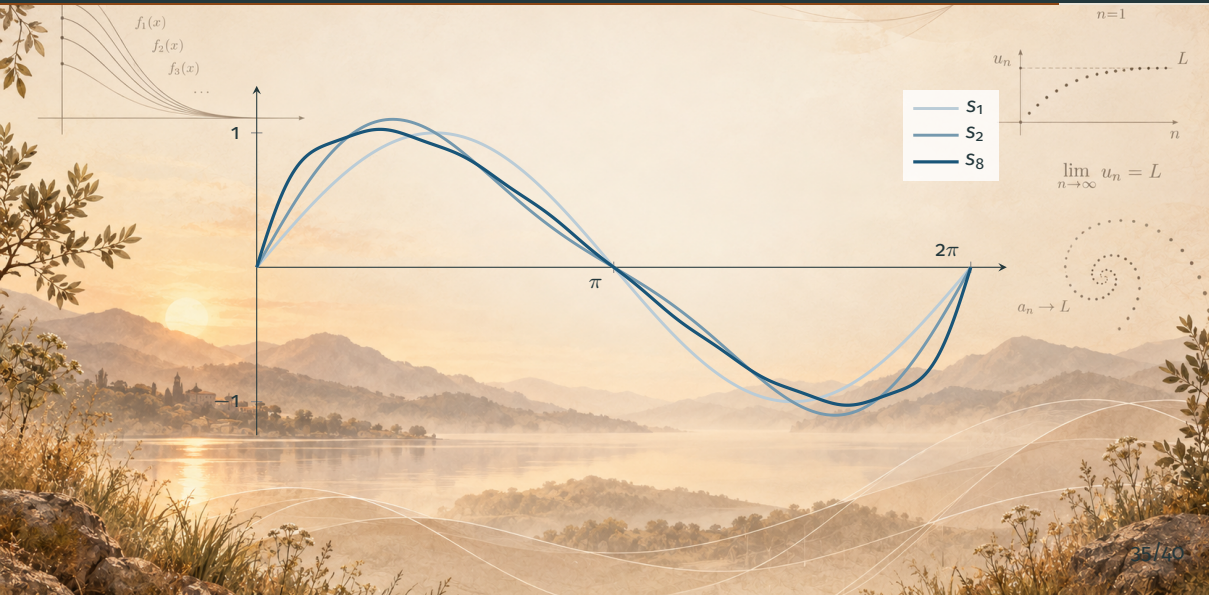
$$\sum_{n=1}^{\infty} \frac{\sin nx}{n^2}, \quad \left| \frac{\sin nx}{n^2} \right| \leq \frac{1}{n^2} = M_n \quad (x \in \mathbb{R}), \quad \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty.$$

$\Rightarrow$ the series converges **uniformly on all of $\mathbb{R}$** .

Error bound, uniform in x :

$$|f(x) - s_N(x)| \leq \sum_{n>N} \frac{1}{n^2} < \sum_{n>N} \frac{1}{n(n-1)} = \boxed{\frac{1}{N}} \quad (x \in \mathbb{R}).$$

Example: Partial Sums of $\sum \frac{\sin nx}{n^2}$



The Converse of the M-Test Fails

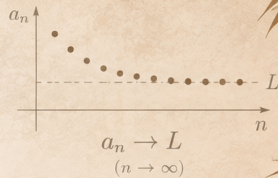
Uniform convergence without a summable bound

Let $f_n(x) = \frac{(-1)^{n+1}}{n}$ (constant functions) on any set E .

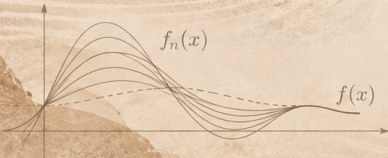
- $\sum f_n$ converges **uniformly**: the partial sums do not depend on x , and $\sum \frac{(-1)^{n+1}}{n}$ converges (Theorem 3.43)
- Any admissible bound has $M_n \geq |f_n(x)| = \frac{1}{n}$, so $\sum M_n = \infty$

Moral: M-test is *sufficient*, not necessary. Cancellation can give uniform convergence.

$$f_n(x) \rightarrow f(x) \\ (n \rightarrow \infty)$$



Summary



$$S_N = \sum_{n=1}^N u_n$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

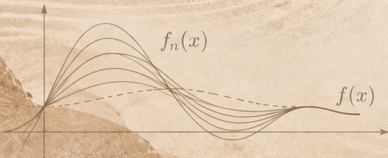
Summary

Key takeaways from this section:

- **Definition 7.7:** $f_n \rightarrow f$ **uniformly** on E : one $N = N(\varepsilon)$ for all x ; f_n stays in the ε -tube

$$a_n \rightarrow L \\ (n \rightarrow \infty)$$

n



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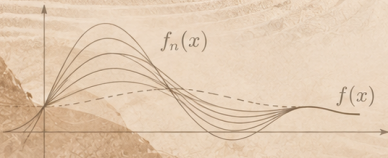
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$$a_n \rightarrow L$$

$(n \rightarrow \infty)$

n



$$S_N = \sum_{n=1}^N u_n$$

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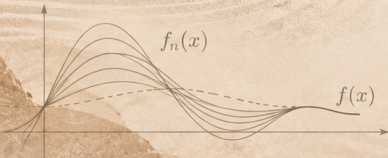
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$$a_n \rightarrow L \\ (n \rightarrow \infty)$$

n

$$S_N = \sum_{n=1}^N u_n$$

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- **Definition 7.7:** $f_n \rightarrow f$ **uniformly** on E : one $N = N(\varepsilon)$ for all x ; f_n stays in the ε -tube
- **Theorem 7.8 (Cauchy):** uniform $\iff |f_n - f_m| \leq \varepsilon$ on E for $m, n \geq N$; **no f needed**
- **Theorem 7.9:** uniform $\iff M_n = \sup_E |f_n - f| \rightarrow 0$; Examples 7.3, 7.6 fail, Example 7.5 passes
- **Theorem 7.10 (M-test):** $|f_n| \leq M_n, \sum M_n < \infty \Rightarrow \sum f_n$ uniform; converse false

Next: uniform limits of continuous functions are **continuous**.

$$a_n \rightarrow L \\ (n \rightarrow \infty)$$

n

$$\sum_{n=1}^N$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$